System Architecture and Data Flow

Detailed project document · Prepared for project completion and review

Project reference information

Item	Known project detail
Project	Health360 Clinical & Healthcare Risk Analysis
Dataset size	5,000 patient/person records
Dataset fields	17 columns; exact field names and definitions must be copied from the source dataset's header row/data dictionary
Analytics tool	Tableau
Visual assets	9 worksheets/visualizations and a dashboard; a Tableau story for Patient Risk & Lifestyle Insights was also part of the project workflow
Repository	https://github.com/en25cs3010918-ctrl/Health360-Healthcare-Risk.git
Website	https://en25cs3010918-ctrl.github.io/Health360-Healthcare-Risk/
Scope caution	This is an educational data-visualization project. It is not a clinically validated diagnostic, prognostic, or treatment system.

Architecture summary

The project workflow consists of a source dataset, a preparation/validation step, a Tableau workbook, worksheet-level analyses, a dashboard, a story, and a website that provides project context and embeds or links to the published views. The GitHub repository is https://github.com/en25cs3010918-ctrl/Health360-Healthcare-Risk.git and the project website is https://en25cs3010918-ctrl.github.io/Health360-Healthcare-Risk/. Confirm that these URLs remain current before submission.

Data flow

1. Acquire the authorized source dataset. 2. Preserve an untouched source copy where permitted. 3. Inspect columns and types. 4. Apply documented cleaning rules. 5. Validate counts and derived values. 6. Connect the cleaned data to Tableau. 7. Build worksheets. 8. assemble the dashboard and story. 9. Publish views using the intended permissions. 10. Embed or link the published views from the website. 11. Test links, filters, and responsive layout. 12. Update documentation and version records.

Component responsibilities

Source data provides the raw records and field definitions. Preparation tools apply cleaning and derived-field rules. Tableau performs the documented analysis and visualization. Tableau Public or the approved Tableau environment serves published views. GitHub stores website source code and project documentation if committed. GitHub Pages hosts the static website when configured. These responsibilities should be adjusted if the actual implementation differs.

Interfaces and failure points

Potential failures include a missing or changed source file, renamed fields, a broken data connection, a calculation mismatch, a private/unpublished Tableau view, a malformed embed URL, browser restrictions, network filtering, or a website deployment that has not completed. Troubleshooting should isolate each layer: open the source, open the workbook, open each published view directly, then test the website embed.

Design decisions

Prefer a documented, reproducible pipeline over undocumented manual changes. Keep workbook calculations understandable and avoid duplicate calculations that disagree. Use public embedding only if the data is approved for public access. Maintain a clear distinction between analytical content and website presentation.

Completion and review record

Prepared by: _____ Date: _____

Reviewed by: _____ Status: Draft / Reviewed / Approved

Document-control note: Update this file when the dataset, workbook, published views, or website changes. Record version/date and keep claims tied to evidence. This detailed document uses known project context and explicitly flags details that must be verified from the actual source files.